



galleys struggled to survive in heavy oceanic seas. By the 16th century, states around Europe's Atlantic coast had developed carracks – vessels with high castles (multi-deck structures) fore and aft, and a combination of square and lateen (triangular) sails. They were effective warships, and large carracks, known as “great ships”, became prestigious status symbols for early 16th-century monarchs.

EVOLVING SHIPS

From the mid-16th century, the galleon, a slimmed-down, faster version of the carrack, was the pivotal warship. The invention of the gun port allowed



large numbers of heavy cannon to be mounted on ships' lower decks, and firing broadsides (from the ship's sides) became the norm. In the 17th century, the galleon evolved into the ship of the line, built to fight in lines together, firing broadsides, and supported by smaller frigates and fireships.

Warships were essential to the states of western and northern Europe as they competed for trade routes and colonies, but permanent navies were expensive to maintain. All countries conscripted armed merchantmen into their navies, and also depended on privateers – licensed pirates who preyed upon foreign states' shipping and colonies.

In the 16th century, Spain had been the dominant naval power, using galleys in the Mediterranean, and galleons in the Atlantic. By the 17th century, the Dutch and English were vying for naval supremacy in a series of large-scale naval battles in the Anglo-Dutch Wars from the 1650s to the 1670s. France, on the other hand, devoted serious resources to naval development only in the reign of Louis XIV, in the 1660s.

The tactics of battles between sailing ships evolved in an ad hoc fashion. When the Spanish sent their Armada against England, in 1588, the smaller English galleons sought to duel with cannon, while the Spanish, their ships packed with soldiers, would have preferred to board: since it was hard to sink a wooden sailing ship with cannon fire, boarding remained a prime tactic. In the 17th century, however, navies formalized a system for exchanging broadsides, but despite this, naval battles remained brutal due to their sheer quantity of firepower.

▲ AT CLOSE QUARTERS

The last naval encounter of the Anglo-Dutch Wars was the Battle of Texel, fought off the coast of the Netherlands, in August 1673. While inconclusive, the battle was contested with savage cannon broadsides and hand-to-hand fighting with cutlasses, daggers, and axes.

KEY EVENTS

1500–1700

■ **1509** The defeat of an Ottoman, Egyptian, and Gujarati (western Indian) fleet by the Portuguese at the Battle of Diu establishes the superiority of European sailing ships in the Indian Ocean.

■ **1511** The English carrack *Mary Rose* is one of the first ships to be built with gun ports.

■ **1571** The Battle of Lepanto takes place. Fought between Christian and Muslim fleets in the Mediterranean, more than 400 oared galleys and over 100,000 men are involved (see pp. 154–55).

■ **1582** The first naval battle in mid-ocean occurs off the Azores – an archipelago in the North Atlantic ocean – with Spanish galleons beating the French and Portuguese.

■ **1588** Spain assembles 130 ships for its Armada to sail to the English Channel; less than half return, with most losses due to storms.

■ **1639** Dutch admiral Maarten Tromp destroys a Spanish fleet at the Battle of the Downs by making effective use of fireships; this marks the end of the era of Spanish naval dominance.

■ **1653** Fighting Instructions drawn up for the British Royal Navy by Admiral Robert Blake (known as the “Father of the Royal Navy”) order captains to enter combat in a disciplined line of battle.

“Every squadron shall endeavour to keep in line with the chief, unless the chief be maimed or otherwise disabled (which God forbid!)”

FIGHTING INSTRUCTIONS ISSUED TO THE ENGLISH FLEET, MARCH 1653